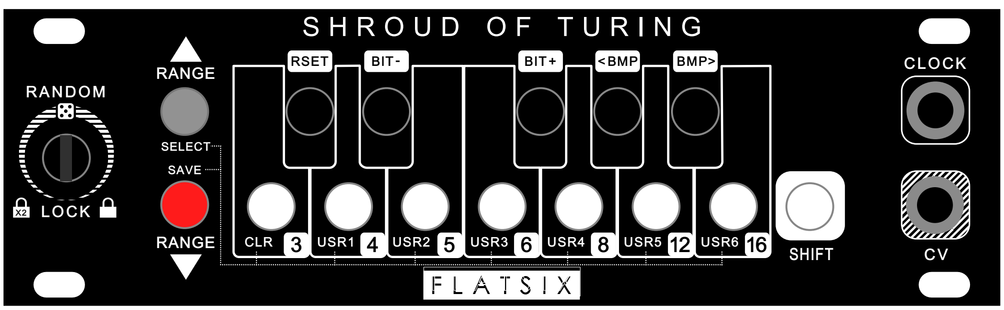
SHROUD OF TURING

USER MANUAL

Firmware Version 1.2.0

FLATSIX
MODULAR



Module Specs

- **Height:** 1U (Intellijel format)
- **Width:** 26HP
- **Depth:** 40mm (“Skiff-friendly”)
- **Current Draw:** 40 mA +12V, 0 mA -12V, 0 mA +5V
- **CV Output:** +0 to +4V Calibrated to 1V/O Standard
- **Gate Input:** +5V clock or gate in

WWW.FLATSIXMODULAR.COM

NOCTURNE ALCHEMY
PLATFORM

FLATSIX
MODULAR

Welcome to the **Nocturne Alchemy Platform** – a flexible eurorack module series that redefines versatility and creativity. Each module within the Nocturne Alchemy Platform shares the same robust Arduino-based hardware, allowing you to effortlessly swap functionalities through our intuitive web firmware loader.

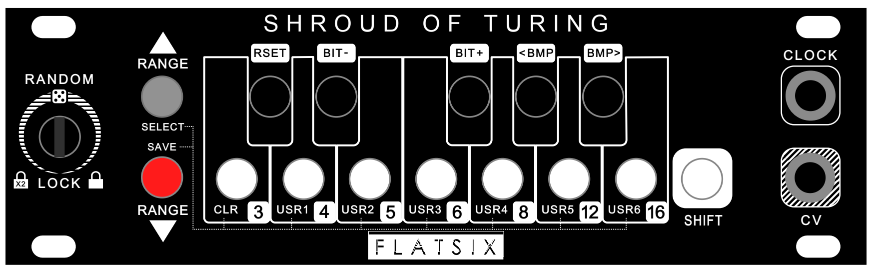
By purchasing one module, you gain access to the full range of available firmwares, including our current creations, **Shroud of Turing**, **Slight of Hand**, **Seventh Summoner**, and the **Arp Of Darkness**, as well as others planned for future release!

This means your module can evolve with your music, with new and exciting functionalities just a simple firmware update away. Embrace the magic of endless possibilities with the Nocturne Alchemy Platform and try some of the other free firmwares today!

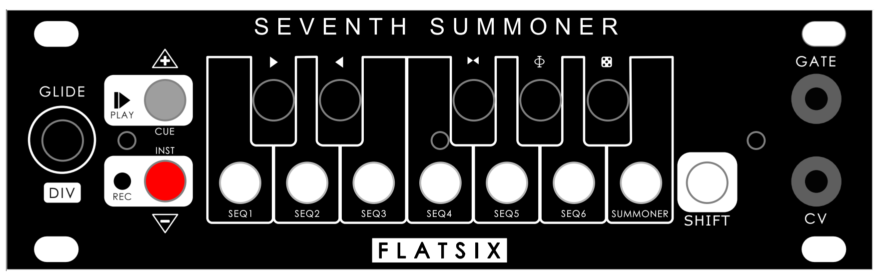
Install FREE firmwares here:

https://flatsixmodular.com/firmware/

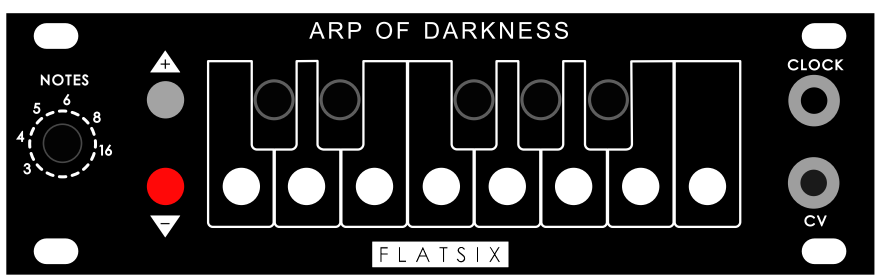
Shroud of Turing — Turing Machine-inspired performance-based random sequencer with quantization, user presets, and scale recall.



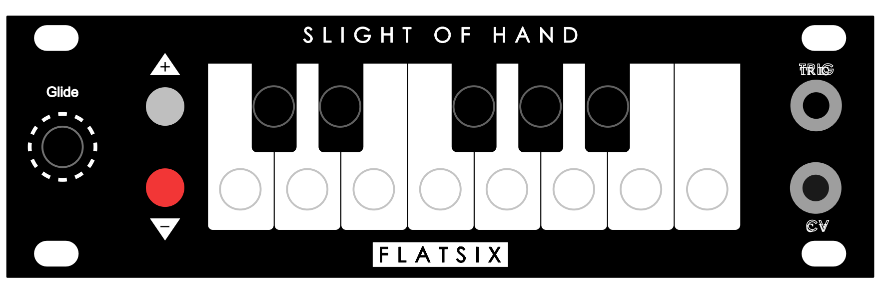
Seventh Summoner — Performance sequencer able to record six 32-note patterns and a meta-sequencer to control them all.



Arp of Darkness — Monophonic, buffer-based arpeggiator with multiple playback modes, state recall, and golden-ratio sequencing.

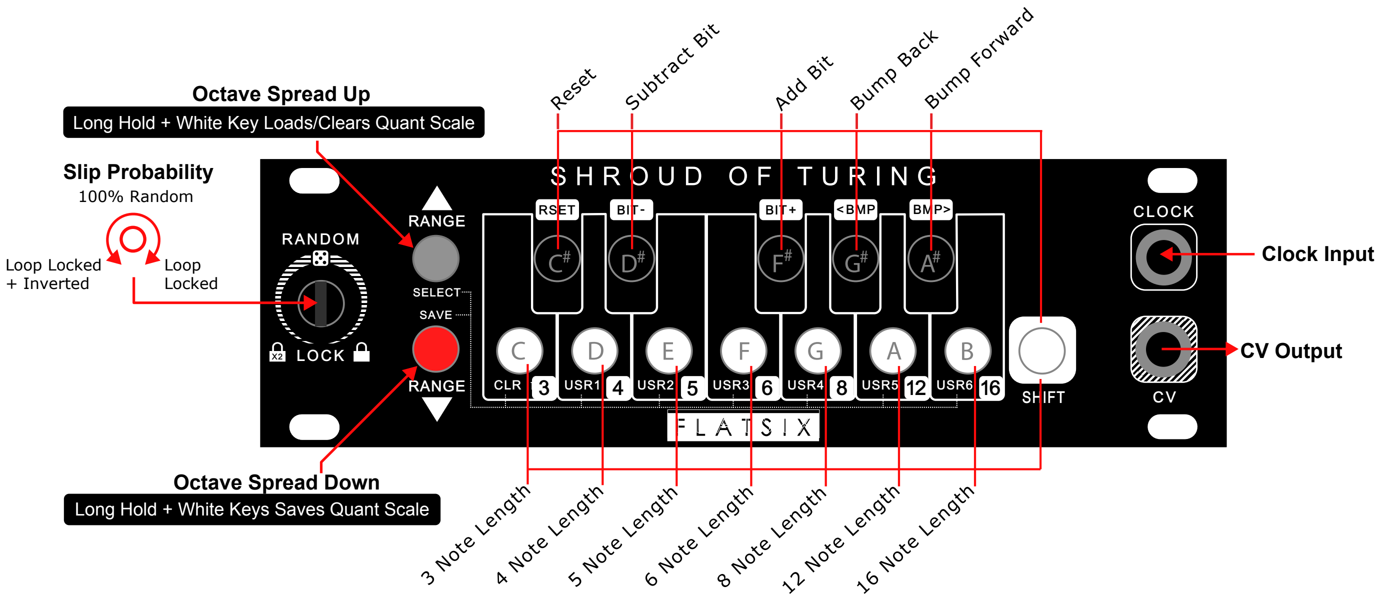


Slight of Hand — 1U, precision manual CV/Gate controller with four octave range and portamento control.



SHROUD OF TURING
CHEAT SHEET

FLATSIX



Main Knob Positions

Position	Mode	Behavior
7 o'clock	DOUBLE	Pattern plays, then inverted (2x length)
8-11 o'clock	SLIP	Pattern slowly evolves (1-100% probability)
12 o'clock	RANDOM	Total chaos (100% random)
1-4 o'clock	SLIP	Pattern slowly evolves (100-1% probability)
5 o'clock	LOCKED	Perfect repeating loop (0% changes)

Key Button Functions

Button	Normal Mode	+ SHIFT
C	Add C to scale	3-step sequence
C#	Add C# to scale	RESET pattern
D	Add D to scale	4-step sequence
D#	Add D# to scale	Clear current bit
E	Add E to scale	5-step sequence
F	Add F to scale	6-step sequence
F#	Add F# to scale	Set current bit
G	Add G to scale	8-step sequence
G#	Add G# to scale	Rotate backward
A	Add A to scale	12-step sequence
A#	Add A# to scale	Rotate forward
B	Add B to scale	16-step sequence

Octave Button Functions

Action	Function
Octave Range Up (tap)	Increase voltage range (1→2→3→4)
Octave Range Down (tap)	Decrease voltage range (4→3→2→1)
Long Hold Octave Up *(Select)* + Note	LOAD scale from slot
Long Hold Octave Down *(Save)* + Note	SAVE scale to slot
Long Hold Octave Up + C	CLEAR scale (unquantized)

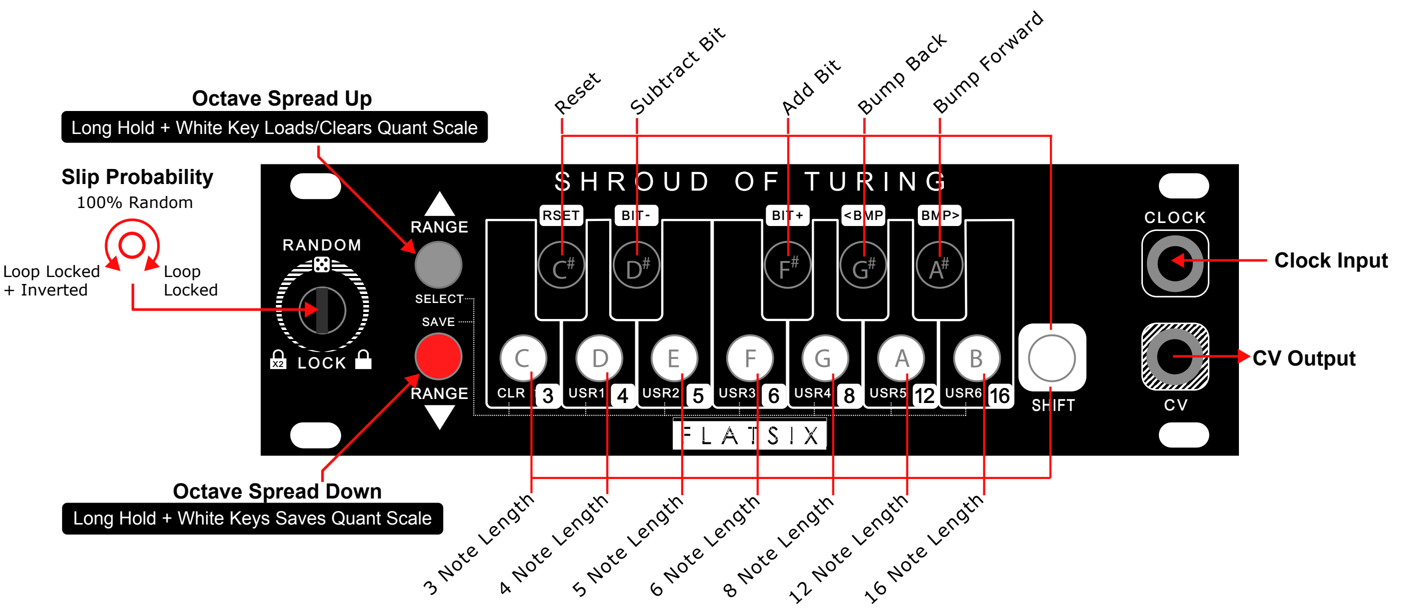
Scale Save/Load Slots

Note Button	Save/Load Slot
Oct Up/Down +D	User Scale Slot 1
Oct Up/Down +E	User Scale Slot 2
Oct Up/Down +F	User Scale Slot 3
Oct Up/Down +G	User Scale Slot 4
Oct Up/Down +A	User Scale Slot 5
Oct Up/Down +B	User Scale Slot 6

SHROUD OF TURING

QUICK START

The Shroud of Turing is a 16-bit perfromance-focused shift register sequencer with musical quantization, inspired by Tom Whitwell's Music Thing Modular Turing Machine. It generates pseudo-random CV sequences that can lock into repeating patterns or evolve over time. What sets it apart is the ability to quantize output to custom scales built from the included keyboard, with six memory slots for saving and recalling your favorite scales in real-time.



Your First Patch

1. **Connect a clock source** (LFO, sequencer gate, etc.) to the CLOCK input
2. **Connect the CV output** to your oscillator's V/Oct input
3. **Turn the main knob** to 12 o'clock (fully random)
4. **Start your clock** — you'll hear random voltages
5. **Press some note buttons** (C, E, G) to create a scale — now it's quantized!
6. **Slowly turn the knob clockwise** toward 5 o'clock
7. **Listen for the sweet spot** — the pattern repeats but occasionally drifts
8. **Lock it** at 5 o'clock when you find a pattern you love
9. **Save your scale —** Long hold Octave Down + D
10. **Load anytime —** Long hold Octave Up + D

FLATSIX
MODULAR

1. Understanding the Shift Register

What Is a Shift Register Sequencer?

The Shroud of Turing is inspired by Tom Whitwell's legendary Music Thing Modular Turing Machine—a pioneering design that popularized shift register sequencing in the Eurorack world. At its heart is a 16-bit shift register—a string of 1s and 0s that rotates continuously, generating a stream of pseudo-random voltages.

Think of it as a loop of binary beads:

```
[1][0][1][1][0][0][1][0][1][1][0][1][0][0][1][1]
 ↑
 Current position
```

On each clock pulse:

1. The pattern shifts one position
2. The last bit wraps around to the front
3. Based on the **probability setting**, that bit might flip (0→1 or 1→0)
4. The current pattern is converted to a voltage
5. **If quantization is active**, the voltage snaps to your scale

The magic is in the probability control—the main knob determines how likely each bit is to change as it cycles through

The 16-Bit Register

The shift register holds 16 bits, but you can choose how many steps make up your active sequence (3-16 steps). Only the rightmost bits matching your step length are used for output.

```
Full 16-bit register:  [1][0][1][1][0][0][1][0][1][1][0][1][0][0][1][1]

8-step sequence uses:                          [1][1][0][1][0][0][1][1]

                                                ^^^^^^^^^^^^^^^^
                                        These 8 bits → voltage output
```

6VI

2. Operating Modes

RANDOM Mode (12 o’clock)

Dead center is a place for you to embrace your inner-chaos.

How it works:
- 100% probability of bit flipping
- Every step is essentially random
- No repeating pattern

Best for: Chaotic textures, noise generation, unpredictable modulation

LOCKED Mode (5 o’clock)

Turn the knob fully clockwise to lock the pattern.

How it works:
- 0% probability of bit flipping
- Pattern repeats perfectly forever
- The shift register keeps rotating, but bits never change

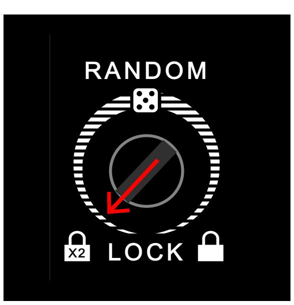
Best for: Finding and keeping a pattern you love, synchronized loops

SLIP Modes (8-11 o’clock and 1-4 o’clock)

The vast middle range of the knob creates SLIP modes—where the pattern slowly evolves.

The Sweet Spot::
- 69% of knob range is dedicated to 1-15% probability
- Patterns repeat but occasionally mutate
- Perfect for evolving sequences that stay musical

Lower SLIP (8-11 o’clock): Probability increases from 1% to 100% **Upper SLIP (1-4 o’clock):** Probability decreases from 100% to 1%



DOUBLE LOCKED Mode (7 o’clock)

Turn the knob fully counter-clockwise to engage **DOUBLE LOCKED** mode.

How it works:
- The feedback bit is always inverted before cycling back
- Creates a pattern that plays normally, then inverted
- Effective sequence length is 2x your setting

Example: An 8-step sequence becomes 16 steps (8 normal + 8 inverted)

Best for: Complex, symmetrical patterns with natural variation

3. Musical Quantization

Performative Quantization

Unlike other shift register sequencers, **Shroud of Turing includes musical quantization through its' built-in keyboard.** This transforms random voltages into actual musical notes in a performace-oriented way..

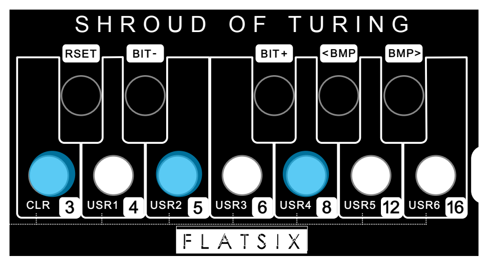
Without Quantization (Empty Scale):
- Default state on boot is unquantized
- Classic shift register behavior
- Full voltage range available
- Output can be any voltage within your current octave-range

With Quantization (Scale Defined):
- Output snaps to nearest note in your scale
- Always in-key, always musical
- Creates melodic sequences instead of random pitches

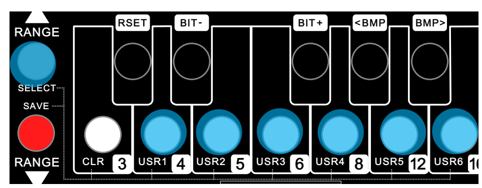
Building Scales

Method 1: Add Notes One at a Time:
Simply press the note buttons to add notes to your scale:
- Press C → C added to scale
- Press E → E added to scale
- Press G → G added to scale
- Result: C major triad (C, E, G)

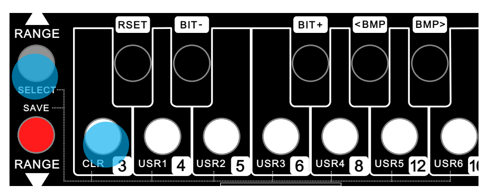
The scale builds as you press buttons. Each press **adds** that note.



Method 2: Load a Saved Scale:
Long hold **Octave Up** (*Select*), then **press D/E/F/G/A/B** to instantly load a saved scale



Clearing the Scale:
Long hold **Octave Up** (*Select*) + C to clear the scale and return to unquantized mode.



How Quantization Works

When a scale is active, the shift register's voltage output is quantized:

1. Shift register generates a raw voltage value
2. That value is mapped to a semitone across your voltage range
3. The semitone is snapped to the nearest note in your scale
4. The calibrated voltage for that note is output

Example with C Major Pentatonic (C, D, E, G, A):
Raw voltage → Would be F → Snaps to E (nearest scale note)
Raw voltage → Would be B → Snaps to A (nearest scale note)

Quantization + Voltage Range

The voltage range setting determines how many octaves your scale spans:

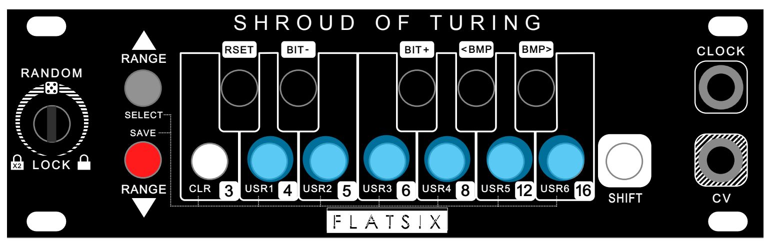
Range	Octaves	Result
1	0-1V	1 octave of your scale
2	0-2V	2 octaves of your scale
3	0-3V	3 octaves of your scale
4	0-4V	4 octaves of your scale

Tip: Smaller ranges with fewer notes create tighter, more focused melodies. Larger ranges with more notes create wider, more dramatic sequences.

---

4. Scale Save/Recall

Why Save Scales?



Building the perfect scale takes time. With **6 user save slots**, you can:

- Store your favorite scales for instant recall
- Switch between scales during performance
- Keep scales across power cycles
- Organize scales by song, mood, or genre

The scale builds as you press buttons. Each press **adds** that note.

Saving a Scale

1. **Build your scale** by pressing note buttons
2. **Long hold Octave Down** (hold for 150ms+)
3. **While still holding**, press one of these buttons:
   - D = User Slot 1
   - E = User Slot 2
   - F = User Slot 3
   - G = User Slot 4
   - A = User Slot 5
   - B = User Slot 6

Example: Save a Pentatonic Scale

1. Press C, D, E, G, A (builds pentatonic)
2. Long hold Octave Down
3. Press D (saves to Slot 1)
4. Done! Scale saved to Slot 1.

Loading a Scale

1. **Long hold Octave Up** (hold for 150ms+)
2. **While still holding**, press slot button (D/E/F/G/A/B)
3. Scale loads instantly!

Example: Load from Slot 1

1. Long hold Octave Up
2. Press D
3. Done! Slot 1's scale is now active.

Clearing the Scale

To return to unquantized mode:

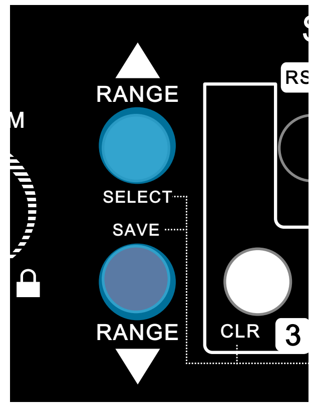
1. Long hold Octave Up (hold for 150ms+)
2. While still holding, press C
3. Scale cleared—back to unquantized voltages

---

5. Voltage Range Control

Adjusting the Range

- Octave Up (tap): Increase range (1→2→3→4 octaves)
- Octave Down (tap): Decrease range (4→3→2→1 octaves)



Musical Applications

Range	Best For
1 octave	Bass lines, focused melodies
2 octaves	Standard melodic range
3 octaves	Wide melodies, arpeggios
4 octaves	Dramatic jumps, full range

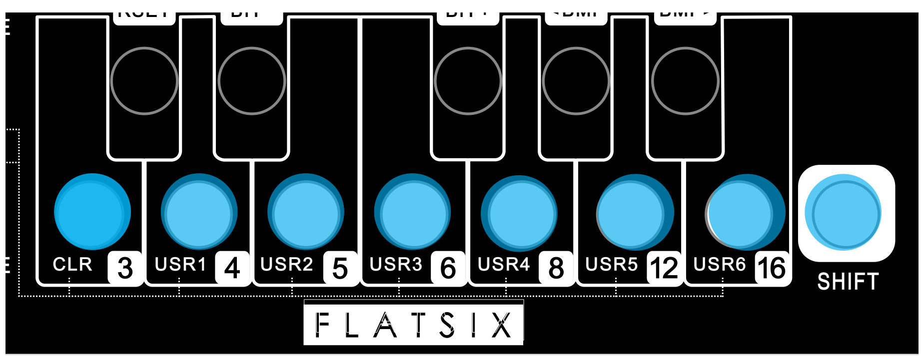
Tip: *Change the range during performance for dynamic shifts—narrow for verses, wide for choruses.*

6. Sequence Length

Available Lengths

The Shroud of Turing has seven diffferent sequence buffer lengths that can be adjusted on the fly. To change the length, simply hold **SHIFT** (High C button) and press:

Button	Length
C	3 steps
D	4 steps
E	5 steps
F	6 steps
G	8 steps (default)
A	12 steps
B	16 steps



Choosing a Length

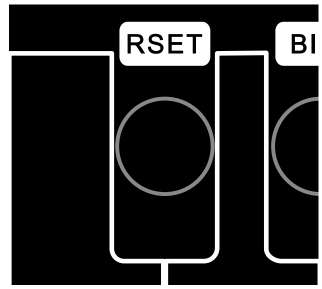
- **3-4 steps**: Tight, hypnotic loops. Great for basslines.
- **5-6 steps**: Odd meters, interesting polyrhythms against 4/4
- **8 steps**: Classic, versatile length
- **12 steps**: Extended phrases, room to breathe
- **16 steps**: Maximum length, complex evolving patterns

Tip: *Shorter sequences lock into groove faster. Longer sequences take more time to find the sweet spot but offer more variation.*

CORE CONCEPTS

6. Pattern Reset

The reset function allows you to restart a **locked** sequence from the first note in the buffer in order to sync it to the start of your song or a downbeat. You can reset when the clock is stopped to start on the first note of the sequence when the clock starts, or you can reset in realtime while playing to sync the first note to whatever beat you please.

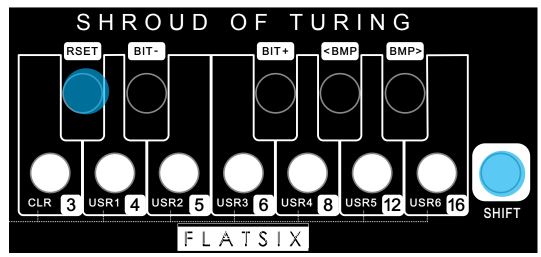


When Does Reset Work?

- LOCKED mode (5 o'clock)
- DOUBLE mode (7 o'clock)
- When the clock is stopped or started

How to Reset

1. Lock your pattern (turn knob to 5 or 7 o'clock)
2. Let it play through at least once
3. Hold SHIFT (High C)
4. Press C#
5. On the next clock pulse, the sequence jumps to step 1



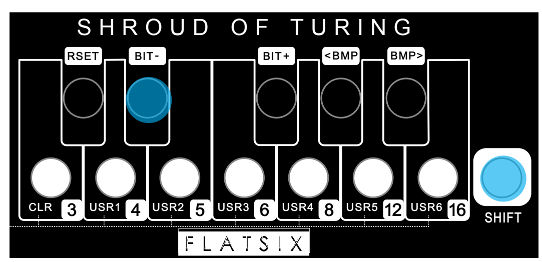
7. Write Functions

The Shroud of Turing includes **manual pattern editing** via the WRITE functions, inspired by the original Music Thing Modular design's WRITE switch.

Clear Bit (SHIFT + D#)

Forces the current bit to 0 (zero voltage contribution).

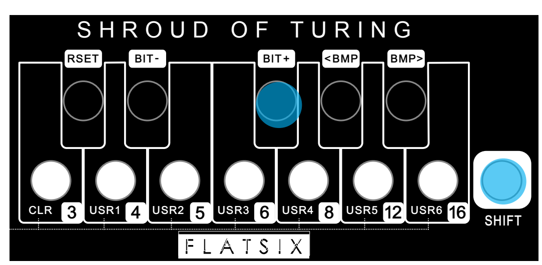
How to use:
1. Hold SHIFT
2. Press and hold D#
3. Each clock pulse clears that step's bit



Set Bit (SHIFT + F#)

Forces the current bit to 1 (maximum voltage contribution for that position).

How to use:
1. Hold SHIFT
2. Press and hold F#
3. Each clock pulse sets that step's bit



8. Pattern Rotation

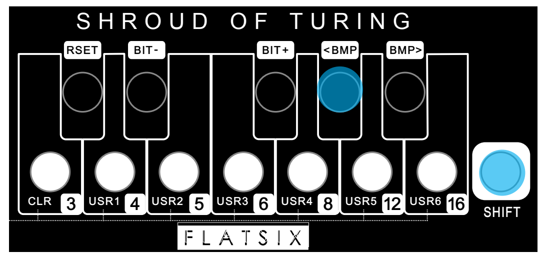
Nudging or "Bumping" the Pattern

Sometimes you find the perfect pattern, but it starts on the wrong beat. Pattern rotation lets you shift the entire pattern forward or backward without changing its content when the pattern is locked.

Rotate Backward - SHIFT + <BMP (G#)

Wraps the first note to the end and moves other notes left once per button press:

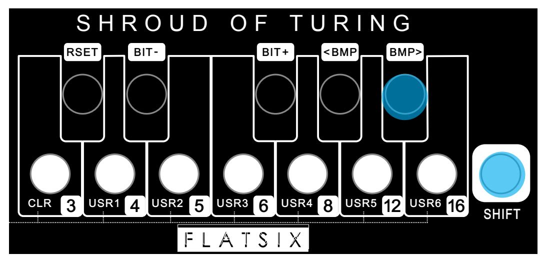
Before: [A][B][C][D]
After: [B][C][D][A]



Rotate Foreward - SHIFT + BMP> (A#)

Wraps the last note to the beginning and moves other notes right once per button press:

Before: [A][B][C][D]
After: [D][A][B][C]



When Does Rotation Work?

- LOCKED mode (5 o'clock)
- DOUBLE mode (7 o'clock)
- When the clock is stopped or started

---

9. Scale Reference

Common Scales to Save

- **Major (7 notes)**: C, D, E, F, G, A, B Happy, bright, positive
- **Natural Minor (7 notes)**: C, D, D#, F, G, G#, A# Dark, moody, sad
- **Pentatonic Major (5 notes)**: C, D, E, G, A Universal, always sounds good
- **Pentatonic Minor (5 notes)**: C, D#, F, G, A# Blues, rock, universal
- **Blues (6 notes)**: C, D#, F, F#, G, A# Bluesy, soulful
- **Harmonic Minor (7 notes)**: C, D, D#, F, G, G#, B Exotic, dramatic, tense
- **Whole Tone (6 notes)**: C, D, E, F#, G#, A# Dreamy, floating, Debussy
- **Diminished (8 notes)**: C, D, D#, F, F#, G#, A, B Tense, symmetrical
- **Chromatic (12 notes)**: All notes Unquantized feel with quantization active

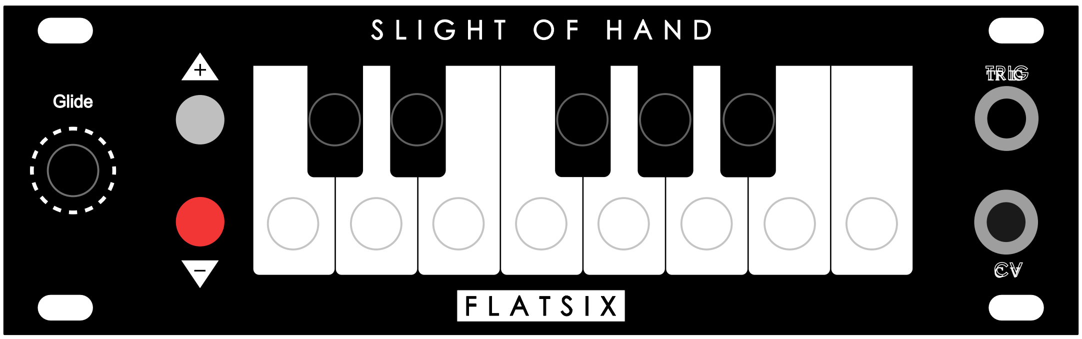
SEVENTH SUMMONER

SLIGHT OF HAND MODE

FLATSIX

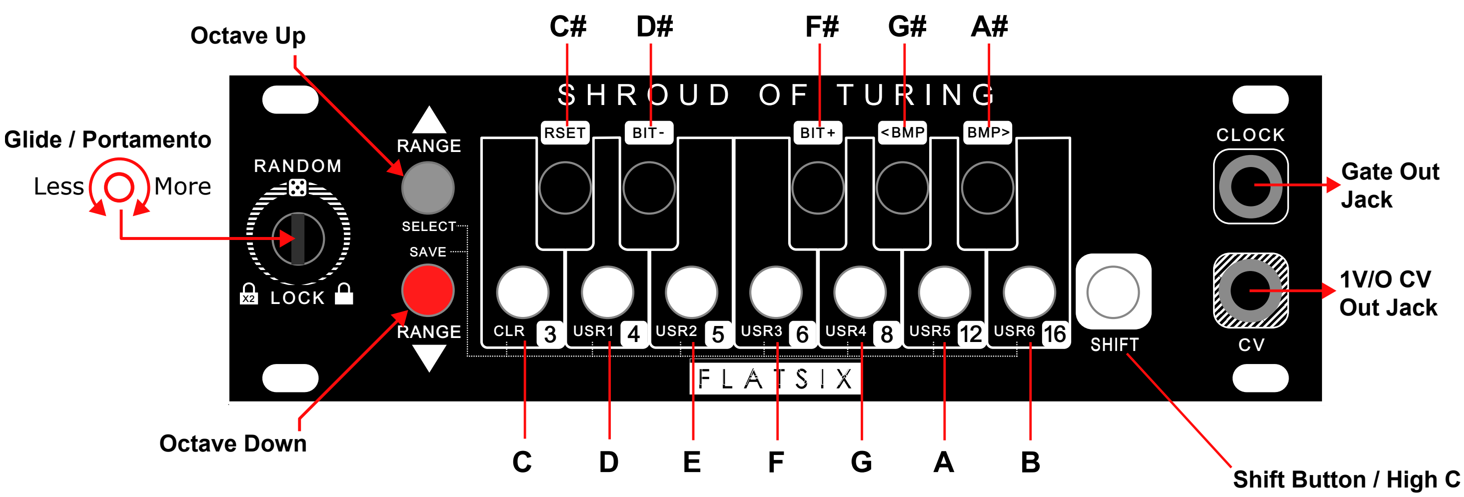
Shape-Shifting Capability: Slight Of Hand Mode

Inspired by studies in lycanthropy, The Shroud Of Turing has gained the ability to shape-shift. Your Shroud Of Turing can boot into an alternate Slight Of Hand mode, functioning identically to the original FlatSix CV keyboard module of the same name. Sometimes you just need a keyboard for that patch, and this mode delivers exactly that without requiring a firmware swap.



In Slight Of Hand Mode:

- The Gate In jack becomes a Gate Out jack, transmitting a 5V gate for the duration of each key press
- The 1V/O CV Out jack outputs the corresponding note voltage while keys are pressed
- The potentiometer adjusts portamento (glide) between notes
- The octave buttons shift the keyboard range up and down
- The SHIFT button functions as a High C note, just like on the original Slight Of Hand



Entering Slight Of Hand Mode:

Hold the SHIFT button while powering on the module (approximately 4 seconds boot time). Upon successful entry into Slight Of Hand mode, the gate output will send 2-3 quick confirmation pulses. You can then release the SHIFT button.

Returning to Shroud Of Turing Mode:

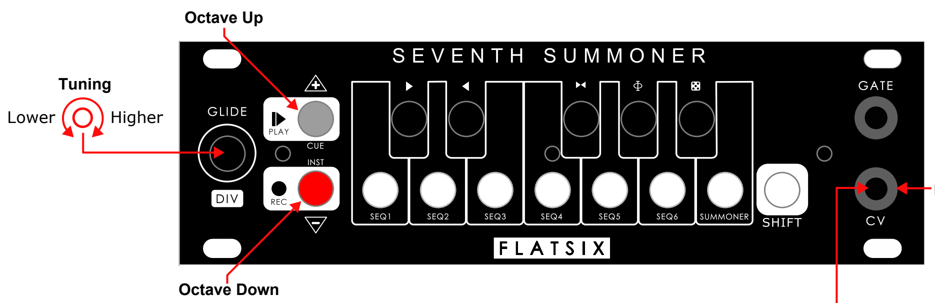
Simply power cycle the module without holding any buttons.

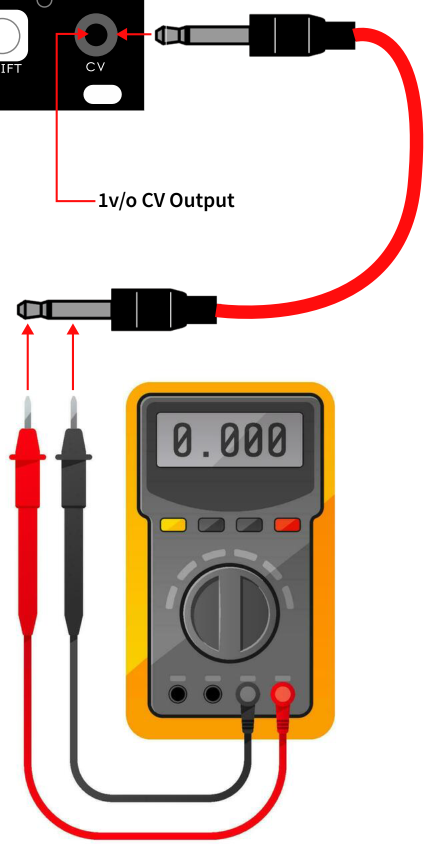
SHROUD OF TURING

CALIBRATION MODE

FLATSIX

The Shroud of Turing just like all Nocturne Alchemy Platform modules features a Calibration Mode that allows users to fine-tune the voltage output for each note, ensuring precise pitch control. This mode is particularly useful for adjusting the module to align with specific musical tuning requirements or to compensate for hardware variances. All modules come precalibrated (every note across 4 octaves) to the closest 1000th of a volt. Most users will never need to calibrate it, but should you choose to, the following procedure is how it is done:





Entering Calibration Mode

Connect multimeter & cable to CV out: Make sure the tip of the cable is connected to signal (red) and the sleeve to the ground (black).

Initiate Calibration Mode: Press and hold both the “Octave Up” + “Octave Down” buttons while powering on the device for about 4 seconds to allow boot to complete. The module indicates entry into Calibration Mode by setting the gate output to HIGH. Then release octave buttons.

Calibration Process

Selecting a Note: Press any note button (except Shift) to select it for calibration. The Shift button acts as a “Shift Key” which enables calibration of the note you selected.

Confirming Selection: Press the Shift button to confirm the selection. This action activates the potentiometer for calibration adjustment.

Adjusting Calibration: Turn the potentiometer to adjust the voltage for the selected note. The adjustment range is limited to one full step above and below the default voltage for finer control.

Saving Calibration: Release the Shift button to save the adjusted voltage. The new calibration value is stored in EEPROM, ensuring persistence across power cycles. Now move to the next note. A best practice is to start at C0, and work your way up through every note across four octaves. This may take some time. (I know because I have handcalibrated every unit sold, so you hopefully won’t have to.)

Special Note Handling

High C (C4): Because the calibration mode uses the high C button on the keyboard as a “shift” button to confirm editing a pitch, the calibration of the final high C4 note is handled differently. When in calibration mode, the octave buttons will allow the keyboard to shift up one additional octave, forcing C4 into the low C key for tuning. No other pitches in that octave will be calibrated, but this allows for calibration of the final high C4.

Exiting Calibration Mode

Exit: Press and hold both the “Octave Up” and “Octave Down” buttons simultaneously for 2 seconds. The module exits Calibration Mode, indicated by the trigger output set to LOW. Now go make some music.

Eurorack Calibration Table For Reference

C	C#	D	D#	E	F	F#	G	G#	A	A#	B
0.000,	0.083,	0.167,	0.250,	0.333,	0.417,	0.500,	0.583,	0.667,	0.750,	0.833,	0.917,
1.000,	1.083,	1.167,	1.250,	1.333,	1.417,	1.500,	1.583,	1.667,	1.750,	1.833,	1.917,
2.000,	2.083,	2.167,	2.250,	2.333,	2.417,	2.500,	2.583,	2.667,	2.750,	2.833,	2.917,
3.000,	3.083,	3.167,	3.250,	3.333,	3.417,	3.500,	3.583,	3.667,	3.750,	3.833,	3.917,
4.000											

Calibration Mode is designed for precision tuning. It is recommended to use a reliable multimeter for accurate calibration.

Resetting to Default Values If for any reason you want to reset all of the calibration values to the module defaults, and start calibrating from scratch, use this function:

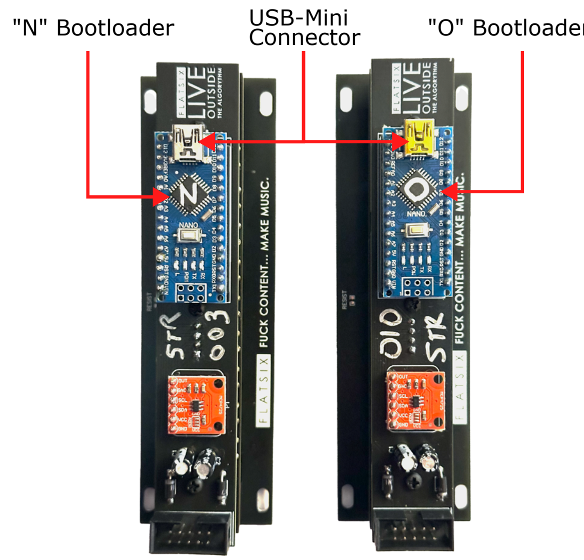
Initiate Reset: While in Calibration Mode, press and hold the “Octave Down” button for 8 seconds without pressing the “Octave Up” button.

Confirmation: The module confirms the reset by blinking the trigger output 6 times. All notes are reset to their default calibration values.

SHROUD OF TURING

FIRMWARE UPDATE

The Shroud Of Turing, just like all the Nocturne Alchemy Platform modules, has an Arduino Nano at the heart of it. Firmware updates can be made using a USB-A to USB Mini cable connected from your computer to the Arduino underneath the module. One can always find the latest firmware version and a link to an online uploader utility at www.flatsixmodular.com/firmware/



Browse to the Firmware Web Uploader Page

The firmware uploader for Arduino-based modules such as the Seventh Summoner only works in Google Chrome on a Mac or PC. You will also need a USB-A to USB Mini cable. If you don't have one laying around, you can find many inexpensive ones on Amazon

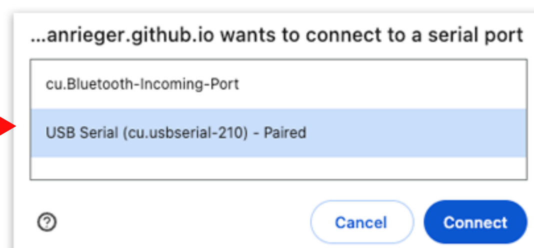
1. Make sure to unplug your module from the Eurorack power cable first.

2. Plug in your USB cable to the mini USB port on the Arp Of Darkness. It's okay to leave the Arduino plugged in to the sockets of the module.

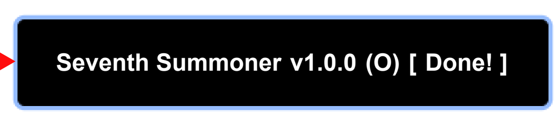
3. The chip next to the USB port will have an "O" or an "N" written on it - On the web uploader page, click the matching red button to upload that firmware.

4. Note: If you don't have an "N" or "O" on the chip, try "O" first, then "N". Guessing won't hurt anything because the uploader will just fail. If it does fail, try the other one (It has to be one or the other and you have a 50/50 chance of getting it right the first time!)

5. Select the "USB Serial" option from the prompt that pops up. It should be highlighted when selected.



6. Click "Connect" and wait while the file uploads until the button says [Done!]



7. That's it. Unplug from the USB cable, plug in the case power, and go make some music.

Legal Disclaimer for FlatSix Modular Products

The use of products manufactured by FlatSix Modular, a DBA of FlatSix Digital, LLC, is entirely at the user's own risk. The products are provided 'as is' without warranty of any kind, either express or implied. Neither FlatSix Modular nor any of its employees, affiliates, or suppliers shall be liable for any direct, indirect, incidental, special, consequential, or exemplary damages, including but not limited to damages for loss of profits, goodwill, use, data, or other intangible losses resulting from the use or the inability to use the products.

By purchasing or using this product, the user acknowledges and agrees to assume all responsibility for all risks associated with all uses of this product, including, but not limited to, the risk of personal injury, damage to property, and/or harm to yourself or others.

Additional Disclaimers:

- FlatSix Modular cannot be held liable for damages resulting from a breakout of a sudden dance party in your house from the sick beats you are dropping with our products. Dance at your own risk.
- Not responsible for any sudden elephant stampedes due to emissions of extreme low frequency sounds. Check your local wildlife regulations before cranking up those bass knobs.
- In no event shall FlatSix Modular be liable for any incidents involving spontaneous time travel, inter-dimensional shifts, or reality distortions that may occur while using our products.
- Use of FlatSix Modular products may not be used as a reason for alien contact, nor can FlatSix Modular be held responsible for any galactic disturbances that follow.
- FlatSix Modular is not responsible for any blisters, calluses, or finger sprains resulting from excessive tweaking, patching, or knob-turning. Modular synthesis is an art and should be approached with caution…and possibly gloves.
- FlatSix Modular accepts no responsibility for any social consequences of becoming too cool or musically enlightened from using our products. If you find yourself suddenly inundated with new friends, handle your popularity at your own risk.
- Be advised that prolonged exposure to modular synths from FlatSix Modular may lead to an insatiable appetite for more modules. FlatSix Modular is not liable for any subsequent financial distress or the need to explain to loved ones why you've converted the dining room into a sound lab.
- FlatSix Modular cannot be held accountable for lost time, be it hours, days or weeks, as you fall down the rabbit hole of patch cables and endless sonic exploration. Users are advised to set alarms, reminders, or have someone check in on them periodically.

Thank you for choosing FlatSix Modular. Remember, with great power comes great responsibility.

FLATSIX DIGITAL

FLATSIX MODULAR